Simulation of the Full Pitt-CoRTEx Detector Stack using Geant4 and NVIDIA OptiX

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PHYS 1378 - Intro to Nuclear and Particle Physics

April 25, 2026

Introduction

Pitt-CoRTEx is a modular detector experiment for educational outreach that tracks cosmic muons. This project will provide computer simulations of particle interactions occurring within the detector stack in an attempt to model its output. The simulations will use the Geant4 software [1] for physics, and the Opticks library [2][3] for photon raytracing (if time permits).

References

[1] "Geant4 Documentation," Geant4, `https://geant4.web.cern.ch/docs/` (accessed Jan. 30, 2026).

[2] Simoncblyth, "Simoncblyth/Opticks: GPU optical photon simulation using NVIDIA OPTIX ray tracing and integrated with Geant4 Simulation," GitHub, `https://github.com/simoncblyth/opticks` (accessed Jan. 30, 2026).

[3] H. Wenzel, S. Jun, K. Genser, and A. Strelchenko, "Integration of OPTICKS1 and GEANT4 (an advanced example: Cats)," Integration of Opticks and Geant4 (an advanced example: CaTS), Feb. 2023. doi:10.2172/1958458